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Original article

Whole-genome sequencing of extensively drug-resistant *Salmonella enterica* serovar Typhi clinical isolates from the Peshawar region of Pakistan



Mah Noor Mumtaz ^a, Muhammad Irfan ^b, Sami Siraj ^c, Aslam Khan ^d, Hizbullah Khan ^{e,f}, Muhammad Imran ^g, Ishtiaq Ahmad Khan ^b, Asifullah Khan ^{a,*}

^a Department of Biochemistry, Abdul Wali Khan University Mardan (AWKUM), Mardan, Pakistan

^b Jamil-ur-Rahman Center for Genome Research, Dr. Panjwani Center for Molecular Medicine and Drug Research (PCMD), International Center for Chemical and Biological Sciences (ICCBS), University of Karachi, Karachi, Pakistan

^c Institute of Pharmaceutical Sciences, Khyber Medical University (KMU), Peshawar, Pakistan

^d Department of Pathology, Medical Teaching Institution MTI, Hayatabad Medical Complex (HMC), Peshawar, Pakistan

^e The Center for Microbes, Development and Health, Key Laboratory of Molecular Virology and Immunology, Shanghai Institute of Immunity and Infection, Chinese Academy of Sciences, Shanghai, China

^f University of Chinese Academy of Sciences, Beijing, China

^g Research center for Advanced Materials Science (RCAMS), Chemistry Department, Faculty of Science, King Khalid University, P.O. Box 9004, Abha 61413, Saudi Arabia

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ABSTRACT

Background: Typhoid fever, caused by *Salmonella enterica* serovar Typhi, is a significant public health concern due to the escalating of antimicrobial resistance (AMR), with limited treatment options for extensively drug-resistant (XDR) *S. Typhi* strains pose a serious threat to disease management and control. This study aimed to investigate the genomic characteristics, epidemiology and AMR genes of XDR *S. Typhi* strains from typhoid fever patients in Pakistan.

Methodology: We assessed 200 patients with enteric fever symptoms, confirming 65 *S. Typhi* cases through culturing and biochemical tests. Subsequent antimicrobial susceptibility testing revealed 40 cases of extensively drug-resistant (XDR) and 25 cases of multi-drug resistance (MDR). Thirteen XDR strains were selected for whole-genome sequencing, to analyze their sequence type, phylogenetics, resistance genes, pathogenicity islands, and plasmid sequences using variety of data analysis resources. Pangenome analysis was conducted for 140 XDR strains, including thirteen in-house and 127 strains reported from other regions of Pakistan, to assess their genetic diversity and functional annotation.

Results: MLST analysis classified all isolates as sequence type 1 (ST-1) with 4.3.1.1. P1 genotype characterization. Prophage and *Salmonella* Pathogenicity Island (SPI) analysis identified intact prophages and eight SPIs involved in *Salmonella*'s invasion and replication within host cells. Genome data analysis revealed numerous AMR genes including *dfra7*, *sul1*, *qnrS1*, *TEM-1*, *Cat1*, and *CTX-M-15*, and SNPs associated with antibiotics resistance. *IncY*, *IncQ1*, *pMAC*, and *pAbTS2* plasmids, conferring antimicrobial resistance, were detected in a few XDR *S. Typhi* strains. Phylogenetic analysis inferred a close epidemiological linkage among XDR strains from different regions of Pakistan. Pangenome was noted closed among these strains and functional annotation highlighted genes related to metabolism and pathogenesis.

Conclusion: This study revealed a uniform genotypic background among XDR *S. Typhi* strains in Pakistan, signifying a persistence transmission of a single, highly antibiotic-resistant clone. The closed pan-genome observed underscores limited genetic diversity and highlights the importance of genomic surveillance for combating drug-resistant typhoid infections.

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* Correspondence to: Department of Biochemistry, Abdul Wali Khan University Mardan, Pakistan.

E-mail address: asif@awkum.edu.pk (A. Khan).

Introduction

Like other infectious diseases, typhoid remains a major public health threat in low-income and middle-income countries [1,2]. This is a severe systemic infection caused primarily by *Salmonella enterica* serovar Typhi and serovar Paratyphi, which are restricted to human hosts [1]. The primary mode of transmission for typhoid fever is direct fecal-oral contact, although recent reports suggest that indirect transmission may also occur due to the bacteria's ability to survive for extended periods in the environment [3]. The global burden of typhoid fever is estimated to be between 11 and 20 million with 128,000–161,000 deaths annually [4]. In 2017, the Global Health Data Exchange reported approximately 116,800 deaths owing to typhoid fever. This includes about 79,000 deaths in South Asia with 6700 from Pakistan [5]. Effective antibiotics play a critical role in treating typhoid; however, the emergence of antibiotic-resistant *S. Typhi* isolates has complicated the clinical management of the disease [6]. These strains have developed multidrug resistance (MDR) against a wide range of antibiotics, and those exhibiting resistance to first-line antibiotics, fluoroquinolones, and third-generation cephalosporins, are classified as extensively drug-resistant strains (XDR) [7].

In recent years, XDR typhoid has posed a significant public health challenge in Pakistan [8]. The resistance of *S. Typhi* to ampicillin, ceftriaxone, co-trimoxazole, and ciprofloxacin has complicated the treatment of enteric fever in the country [9]. There are limited therapeutic options for MDR and fluoroquinolone-resistant strains in countries with high disease burdens, including Pakistan. Several factors have contributed significantly to the emergence of antibiotic resistance in Pakistan, including the use of monotherapy, inadequate guidelines for antibiotic usage, cessation of antibiotic therapy by patients, and prescription without appropriate biochemical testing [7, 10, 11].

The emergence of MDR and XDR *S. Typhi* resulted in a significant socioeconomic burden, particularly impacting individuals from poor economic backgrounds who typically suffer from typhoid fever and the treatment has relatively become expensive. Currently, third-generation cephalosporins, including the most commonly prescribed ceftriaxone, have become the treatment option [12]. However, the first case of ceftriaxone-resistant *Salmonella* XDR typhoid in Pakistan was reported in 2016 from Hyderabad city, situated in south of Pakistan. The World Health Organization (WHO) reported more than 5200 XDR typhoid cases from Southern Pakistan in 2016 [7]. During the emergence of the first outbreak in 2017, more than 17,000 XDR *S. Typhi* cases were reported in the Sindh province of Pakistan. Additionally, the phenotypic and genotypic characterization of 27 XDR *S. Typhi* isolates was reported in 2020 from patients attending Jinnah Hospital in Lahore, Punjab province of Pakistan. The precise burden is not known; however, active surveillance studies from Karachi city of Pakistan reported 4.7 cases of typhoid per 1000 cases per year [13].

Peshawar is the capital of the Khyber Pakhtunkhwa province of Pakistan, located at the northwestern border close to Afghanistan, with a population size of approximately 2.3 million. A recently published study reported high-level drug resistance and the emergence of XDR strains in Peshawar. According to the study, the *S. Typhi* isolates from Peshawar displayed complete resistance, up to 100%, to chloramphenicol, ampicillin, ciprofloxacin, ceftriaxone, and cefixime and additionally exhibited 93% resistance to co-trimoxazole [14]. This report unveils a concerning scenario of *S. Typhi* high-level drug resistance in Peshawar. With the advent of whole genomic sequencing (WGS) technology, WGS-guided epidemiological investigation has become popular in pathogen tracing, outbreak investigation, and drug resistance profiling.

To the best of the authors' knowledge, no previous study has explored the genomic perspective of drug-resistant *S. Typhi* isolates

from Peshawar and the neighboring northwestern region of Pakistan. Therefore, the current study was pursued to understand the genomic characteristics and drug resistance patterns of *S. Typhi* isolates from the Peshawar region of Pakistan.

Materials and methods

Ethical approval, bacterial isolates identification, and antimicrobial susceptibility testing

The current study protocol was approved by the institutional ethical review committee under reference number AWKUM/Biochem/eth-comm/554, in accordance with the 1975 Helsinki guidelines. The study enrolled a total of 200 patients with clinical symptoms of fever, fatigue, headache, nausea, abdominal pain, constipation, and diarrhea and sought medical attention at Hayatabad Medical Complex (HMC), Peshawar between October 2020 and February 2022. Prior to sample collection, written consent forms were obtained from each patient or their guardian.

Whole blood samples were collected from the registered patients admitted at the HMC, Peshawar. Samples were examined in the clinical microbiology laboratory for bacterial culture, identification, and antimicrobial susceptibility testing (AST). The blood samples were first inoculated in BACTEC vials containing growth media optimized for the growth of *S. Typhi*, along with identification markers to facilitate bacterial growth detection. These vials were incubated at 35–37 °C in the BACTEC 9240 blood culture automated system (Becton Dickinson, USA), that provides an ideal environment for bacterial growth. The incubation was performed for approximately 72 h to allow sufficient *S. Typhi* proliferation. Samples showing positive bacterial growth were sub-cultured on MacConkey agar plates at 37 °C for 24 h to isolate the bacteria. The isolated colonies were subjected to biochemical identification by triple sugar iron (TSI) and indole tests. Additional confirmation of *S. Typhi* colonies was performed via Analytical Profile Index (API) 20E test (BioMérieux) [15] and agglutination with genus and serotype-specific antisera (*Salmonella* O antiserum) [16]. The isolated colonies were subjected to AST assay via Kirby-Bauer disk diffusion method [17], following the guidelines outlined by the Clinical & Laboratory Standards Institute (CLSI) in 2021 [18]. The antibiotics, including amoxicillin, chloramphenicol, cotrimoxazole, ciprofloxacin, meropenem, imipenem, cefotaxime, ceftriaxone, and azithromycin were acquired from Oxoid (UK) to perform susceptibility tests.

DNA extraction and whole genome sequencing

Genomic DNA was isolated from the isolated colonies of XDR *S. Typhi* strains using DNA extraction kit (catalog number 11858874001; Roche Basel, Switzerland). The quality of the isolated DNA was evaluated by Agarose gel electrophoresis, and the concentration of DNA was assessed using a Quantus™ Fluorimeter. High-quality DNA samples from thirteen XDR *S. Typhi* strains were processed for paired-end sequencing using HiSeq Illumina platform.

Bioinformatics analysis

The quality assessment of short read paired-end sequenced data was assessed by FASTQC [19]. Adapter sequences and low-quality sequence-reads were filtered using Fastp tool [20]. The de novo genome assemblies were constructed from good quality sequence-reads using Unicycler [21] and SPAdes [22]. The quality of the assembled data was further improved using Pilon package [23]. The quality of the draft genome assemblies was evaluated by QUAST v5.0.2 [24]. Functional annotation of the assembled genome was performed using Prokka v1.14 [25] and NCBI Prokaryotic Genome

Annotation Pipeline PGAP v6.4 [26], which provide faster and more reliable annotation of bacterial genomic DNA sequences.

Phylogenetic analysis

The phylogenetic analysis, of all inhouse sequenced *S. Typhi* along with the reference CT18 strain (accession ID# AL513382.1), Lahore (Punjab) and Hyderabad (Sindh), was performed using REALPHY <https://realphy.unibas.ch/realphy/>. The previously reported sequenced data of XDR *S. Typhi* strains from Punjab [13] and Sindh [7] provinces of Pakistan was retrieved from Pathogen watch resource [27] and the phylogenetic relationship among the all strains was visualized using Figtree v1. 4.4 [28].

Sequence type identification

Multilocus sequence typing was performed using “MLST” v2.17.5 (<https://github.com/tseemann/mlst>, accessed on 1 November 2020) to identify the sequence types (ST) of each sample [29]. MLST is a well-established method extensively used in epidemiological studies to investigate the clonal lineages of bacteria. It provides a reliable and unambiguous procedure for characterizing bacterial isolates by examining their distinctive allelic features derived from the DNA sequences of seven housekeeping genes. Additionally, the genotype information of the samples was identified from their WGS data using Genotyphi scheme of the Pathogen Watch resource [27]. PHASTER and SPIFinder 2.0 resources were utilized at default parameters to identify prophage sequences and Salmonella Pathogenicity Islands, respectively [30].

Antimicrobial resistance, virulence genes, and plasmid identification

The antimicrobial resistance profile of samples from their WGS data was identified using ResFinder v4.1 [31]. Antimicrobial resistance (AMR) gene families and resistance mechanisms were identified using the Resistance Gene Identifier (RGI), i.e. based on the comprehensive antibiotic resistance database (CARD) [32]. The virulence-associated genes were identified and annotated using the Virulence Factor Database (VFDB) at <http://www.mgc.ac.cn/VFs/>, accessed on 16 February 2021 [33]. The high quality sequence-reads data of the samples was separately subjected to de novo assembly using plasmidSPAdes tool [34] to characterize plasmid sequences. The assembled contigs were screened against the Plasmid Sequence Database (PLSDB) [35] using the BLASTn tool by following a threshold of 100% sequence identity, 80% query coverage, and a maximum E-value of 0.0001 [36]. The PLSDB resource contains 13,789 plasmid sequences with records collected from the NCBI database [35].

Pan-genome construction

We conducted the pangenome analysis using the Bacterial Pan Genome Analysis Pipeline (BPGA) [37] with genomic data from 140 XDR *S. Typhi* strains. This dataset comprises data from 13 recently sequenced in-house strains, in addition to previously reported sequences from Lahore (Punjab) (n = 27) and Hyderabad (Sindh) (n = 100) samples. Annotated genomes were utilized to construct the pangenome, encompassing the complete set of genes found in all *S. Typhi* strains. The pan-genome functional analysis module of BPGA was used to assign Cluster of Orthologous Genes (COG) [38,39] and Kyoto Encyclopedia of Genes and Genomes (KEGG) [40] classes to the core, accessory, and unique gene families. The empirical power law equation $f(x) = a \cdot x^b$ and exponential equation $f(x) = c \cdot e^{(d \cdot x)}$ were followed for extrapolation of the pan and core genome curves respectively.

Species-specific gene families were identified by determining the exclusive presence or absence of genes and gene families. Evolutionary analysis based on concatenated gene alignments and binary (presence/absence) pan-matrix was conducted with Neighbor-joining (NJ) approach. The functional characteristics of the pan and core-genome was investigated using COG distribution and KEGG pathway analysis.

Data availability

The FASTQ files of the raw sequence reads of all the samples are available in the NCBI Sequence Read Archive (SRA) database under Bio Project accession no PRJNA832906. The assembled data are deposited at Genbank NCBI under Bio Project no PRJNA839213. The accession IDs of the all thirteen sequenced strains include; JAMZNY000000000, JAPDOU000000000, JAMZNX000000000, JAMZNV000000000, JAMZNV000000000, JAQNS000000000, JAPDOV000000000, JAPDOW000000000, JAPDOX000000000, JAPDOY000000000, JAPDOZ000000000, JAPDPA000000000, and JAPDPB000000000.

Results and discussion

S. Typhi culturing and antimicrobial susceptibility

A total of 200 patients (120 males, 80 females) suffering from fever were registered at the HMC hospital of the Peshawar, during October 2020 till February 2022. Among these, *S. Typhi* strains were confirmed for 65 patients based on biochemical tests, including TSI, indole, and H₂S production (Fig. S1–S2). Additional confirmation of the species was made via Analytical profile index API 20E test (BioMérieux) (Fig. S3) and agglutination with genus and serotype-specific antisera (Salmonella O antiserum). AST assay identified 40 XDR and 25 MDR strains in all 65 *S. Typhi* cases. The XDR strains exhibit resistance to cell wall synthesis inhibitors (amoxicillin, cefotaxime, and ceftriaxone), protein synthesis inhibitor (chloramphenicol), folate inhibitor (co-trimoxazole) and DNA synthesis inhibitors (ciprofloxacin). However, these isolates were found susceptible to azithromycin, meropenem and imipenem. All XDR strains exhibited a more or less similar AST pattern (Table 1; Fig. S4).

Whole genome sequencing

Thirteen XDR *S. Typhi* strains were subjected to complete genome sequencing using the Illumina HiSeq platform. The paired-end reads data with a read length of 150 base pairs (bp) was generated for each XDR sample. Approximately 6 GB genomic data was generated for each sample. About 2% low-quality sequenced reads were filtered out from each sample to ensure data integrity and reliability during the quality filtration steps. The de novo genome assemblies of all the isolates were constructed using Unicycler and SPAdes followed by further refinement using pilon package. The number of assembled contig sequences in different samples were ranges from 59 to 71. The average size of each sample genome was contained of 4754267 ± 26506 nucleotide bases. The GC content of the assembled genome was noted as $52.31 \pm 0.22\%$. The functional annotation of all genome revealed 4443 ± 615 protein coding sequences. The detailed information of each assembled genome along with functional annotation is shown in Table 2.

Isolates sequence type, prophages and SPI characterization

MLST characterises *S. Typhi* sequence type based on genetic polymorphisms in *aroC*, *dnaN*, *hemD*, *hisD*, *purE*, *sucA*, and *thrA* house-keeping genes [29]. The WGS data of the samples generated in the current study covered all these seven gene sequences and

Table 1
Antimicrobial susceptibility profile of *S. Typhi* clinical isolates acquired from Hayatabad-Peshawar, Pakistan.

Name of Antibiotics (OXOID)	Conc. (µg)	Susceptible (mm)	Intermediate (mm)	Resistance (mm)	Antimicrobial susceptibility												
					XDR1	XDR2	XDR3	XDR4	XDR5	XDR6	XDR7	XDR8	XDR9	XDR10	XDR11	XDR12	XDR13
amoxicillin (AML)	10	≥ 18	14–17	≤ 13	R	R	R	R	R	R	R	R	R	R	R	R	R
trimethoprim-sulfamethoxazole (SXT)	20	≥ 16	11–15	≤ 10	R	R	R	R	R	R	R	R	R	R	R	R	R
chloramphenicol (C)	30	≥ 18	13–17	≤ 12	R	R	R	R	R	R	R	R	R	R	R	R	R
cefotaxime (CTX)	30	≥ 26	23–25	≤ 22	R	–	R	R	R	–	–	–	–	–	–	–	–
ciprofloxacin (CIP)	5	≥ 26	22–25	≤ 21	R	R	R	R	R	R	R	R	R	R	R	R	R
azithromycin (AZM)	15	≥ 13	–	≤ 12	S	S	S	S	S	S	S	S	S	S	S	S	S
ceftriaxone (CRO)	30	≥ 23	20–22	≤ 19	–	R	–	–	–	R	R	R	R	R	R	R	R
meropenem (MEM)	10	≥ 23	20–22	≤ 19	–	S	–	–	–	S	S	S	S	S	S	S	S
imipenem (IPM)	10	≥ 23	20–22	≤ 19	–	S	–	–	–	S	S	S	S	S	S	S	S

I = Resistance, I= Intermediated, S = Susceptible

R = Resistance, I = Intermediate, S = Susceptible

provided an opportunity to characterize the isolate sequence type (ST) via the MLST scheme. All the samples were identified as ST1 and belong to 4.3.1.1. P1 genotype based on established Genotypi scheme [41]. The same molecular typing feature has been reported previously for the XDR *S. Typhi* isolates from Sindh and Punjab provinces of Pakistan emerged during 2016 and 2019 regional outbreaks respectively (7, 13). ST1 and ST3 are single-locus variants of ST2, predicted to be the founder strains of *S. Typhi*. The ST1 and ST2 sequence types of *S. Typhi* are predominantly circulating in Eurasia, South America, and Africa [42]. Several studies have reported that ST2 might circulate worldwide, whereas ST1 persists mainly in the Indian subcontinent, including Pakistan [43].

Different types of prophage sequences were detected in WGS data of all 13 XDR *S. Typhi* (Table 3). A total of five intact prophages were detected. Among these, the Salmon_SEN34 and Salmon_Fels_2 prophages were identified among all isolates WGS sequences. However, the PHAGE_Escher_pro483 was identified specifically in case of XDR12 strain. Many other questionable (i.e. predicted with a 70–90 score) and incomplete (score < 70) phage replicons were also detected in XDR strains (Table S1).

SPIFinder identified that all the newly sequenced XDR samples share 8 *Salmonella* pathogenicity islands, i.e. SPI-1, SPI-2, SPI-3, SPI-5, SPI-6, SPI-8, SPI-9 and SPI-10. The SPI-1 and SPI-2 are the well-studied SPIs and reported critical for *Salmonella*'s ability to invade and replicate within host cells [44]. SPI-3 contains genes that enable *Salmonella* to evade the host immune response [45], while SPI-5 is involved in the transport and utilization of nutrients within host cells. SPI-6 formerly known as *Salmonella enterica* centisome 7 island [SCI] encoding proteins involve in T6SS that reported to play pivotal role in *Salmonella* virulence and pathogenesis. SPI-9 is involved in the transport and metabolism of specific nutrients that are important for *Salmonella* survival within the host. SPI-8 is a 6.8 kb region located adjacent to the *pheV* tRNA gene and SPI-10 is a large insertion of 32.8 kb located at tRNA *leuX* in the *S. Typhi* genome and play a role in *S. Typhi* host specificity [46]. SPI-4, was detected in 10 XDR strains. The genes clustered in SPI-4 encode virulence factors associated with different stages of enteric infection. Additionally, SPI-7 was detected in 11 XDR Peshawar strains, which is a relatively large genomic region consists of several genes involve in *S. Typhi* virulence [47] (File S1).

Drug resistant genes, SNPs and virulence factors identification

The CARD database have catalogued the antibiotics resistance genes and chromosomal mutations conferring resistance to different antibiotics. The protein sequence homology based scanning using resistance gene identifier method of CARD database identified a total of 37 AMR genes among inhouse thirteen sequenced strains that conferring resistance to specific antibiotics. Of these seven genes *TEM-1*, *CTX-M-15*, *QnrS1*, *catI*, *qacEdelta1*, *sul1* and *sul2* were predicted using perfect criteria while 31 genes were predicted using strict RGI criteria. Of these AMR genes, fifteen involve in antibiotic efflux resistance mechanism, eleven are involved in antibiotic target alteration and six genes play role in antibiotic inactivation. The detailed of AMR profile of each genome is shown in (File S2). Six SNPs, including D350N, S357N, E350Q, E448K, R234F, and S83F were identified in the sequenced strain playing substantial role in the antibiotic resistance mechanism (Table 4). Five SNPs, including E350Q, E448K, S357N, D350N, R234F were identified for XDR *S. Typhi* Peshawar samples, however these SNPs have not been reported before for *S. Typhi*. Additionally, a S83F SNP in *gyrA* gene was identified for XDR Peshawar strains that has been reported in association with fluoroquinolones resistance in *Salmonella enterica*. Two SNPs, i.e. E350Q, and E448K in *UhpT* and *GlpT* genes, respectively, reported to confer resistance against fosfomycin. Two penicillin-binding protein-3 (PBP3) SNPs, S357N, and D350N, reported to

Table 2
Genome characteristics and Annotation features of Hayatabad-Peshawar *S. Typhi* XDR isolates.

	XDR1	XDR2	XDR3	XDR4	XDR5	XDR6	XDR7	XDR8	XDR9	XDR10	XDR11	XDR12	XDR13
Genome size (mbp)	4.75	4.78	4.74	4.74	4.75	4.7	4.75	4.75	4.8	4.81	4.75	4.75	4.75
Number of Reads (M)	13.26	9.12	21.15	18.07	20.91	10.04	13.14	10.55	10.11	10.84	9.67	10.37	13.44
No of Contigs	62	66	65	68	62	67	61	59	68	71	60	66	64
GC %	52.48	52.32	52.13	52.12	52.06	52.85	52.28	52.37	52	52.34	52.35	52.38	52.38
N50	204168	174568	197822	174566	204169	161070	204169	204168	204171	185188	204169	201842	201842
L50	9	9	10	10	9	9	9	9	9	9	9	9	9
Largest Contig (bp)	488731	488731	392625	392625	488731	488731	488731	488731	488731	488731	488731	488731	488731
Total Genes	4666	4699	4660	4663	6936	4625	4665	4664	4706	4737	4662	4666	4663
Coding	4584	4617	4583	4586	6833	4552	4583	4582	4622	4655	4580	4584	4581
Sequences (CDSs)													
CDSs (protein coding)	4265	4292	4264	4269	6488	4242	4266	4262	4294	4328	4263	4266	4262
RNA genes	82	82	77	77	103	73	82	82	84	82	82	82	82
rRNAs genes	4	4	2	2	19	1	4	4	6	4	4	4	4
tRNAs genes	69	69	66	66	75	63	69	69	69	69	69	69	69
ncRNAs genes	9	9	9	9	9	9	9	9	9	9	9	9	9
Pseudo Genes	319	325	319	317	346	310	317	320	328	327	317	318	319
MLST	1	1	1	1	1	1	1	1	1	1	1	1	1
Plasmids	pMAC, pAbTS2	IncQ1	pMAC, pAbTS2	pMAC, pAbTS2	–	–	–	–	IncQ1	IncY	–	–	–

Table 3
The intact prophages identified in genome sequences of Hayatabad-Peshawar XDR *S. Typhi* isolates.

Predicted Prophages	XDR1	XDR2	XDR3	XDR4	XDR5	XDR6	XDR7	XDR8	XDR9	XDR10	XDR11	XDR12	XDR13
PHAGE_Salmon_SEN34	+	+	+	+	+	+	+	+	+	+	+	+	+
PHAGE_Salmon_RE_2010	+	++	++	–	++	++	++	++	++	++	++	+	++
PHAGE_Salmon_Fels_2	+	+	++	++	++	+	+	++	+	++	+	++	++
PHAGE_Burkho_phiE255	+	–	–	–	–	–	+	+	–	+	+	–	–
PHAGE_Escher_pro_483	–	–	–	–	–	–	–	–	–	–	–	+	–

+, One copy existed; ++, two copies existed; –, absent

Table 4
Chromosomal point mutations reported in association with different antibiotics resistance determining regions.

AMR gene	AMR SNPs	Resistance Type	Frequency (%)
UhpT	E350Q	fosfomycin	95.03%
PBP3	D350N, S357N	cephalosporin, cephamycin, penam	51.85%
GlpT	E448K	fosfomycin	96.02%
gyrA	S83F	fluoroquinolones	99.54%
EF-Tu	R234F	elfamycin	99.41%

confer resistance to beta-lactam antibiotics (i.e. cephalosporin, cephamycin, penam), were also identified for XDR Peshawar strains [48,49]. Besides, R234F SNP that reported to confer pulvomycin resistance [50] was also identified in case of newly sequenced 12 XDR *S. Typhi* samples from Peshawar except XDR6 strain. (Table 4; File S2).

Identification of virulence factors is essential for estimating the pathogenicity of a pathogenic isolates. The analysis of virulence genes annotation identified cytolethal distending toxin B, adherence determinants, secretion systems, and other virulence-relevant genes

among all the 13 XDR isolates. The isolates were found to code for major and minor curli subunits, curli fibers/thin aggregative fimbriae and curli production assembly/transport component. Additionally, the isolates were found to be capable to code for cytolethal distending toxin subunit-B homolog and antimicrobial peptides resistant proteins. The *Salmonella* pathogenicity island 1 and pathogenicity island 2 coding for type III secretion systems were annotated in cases of all the Peshawar-XDR samples. Besides, the isolates were found to code Type-1 fimbriae, Vi polysaccharide export protein, and bovine colonization factors (Table 5; File S3).

Phylogenetic analysis

Phylogenetic analysis were pursued to determine the genomic epidemiological linkage of Peshawar XDR strains with previously reported XDR *S. Typhi* samples from Pakistan. A dataset of 127 *S. Typhi* strains genomes reported from Punjab and Sindh region of Pakistan, thirteen inhouse newly sequenced XDR strains from Peshawar, KPK and one reference CT18 strain was compiled and analysed. The phylogenetic analysis identified that the newly sequenced Peshawar-XDR isolates intermingled and shared a clade

Table 5
The virulence genes shared among Hayatabad-Peshawar XDR *S. Typhi* isolates.

Virulence category	Genes
Fimbrial adherence determinants	<i>csgA, csgB, csgC, csgD, csgE, csgF, csgG, bcfA, bcfB, bcfC, bcfD, bcfG, fimC, fimD, fimF, fimI</i>
Macrophage-inducible genes	<i>mig-14</i>
Magnesium uptake	<i>mgcC, mgcB</i>
Non-fimbrial adherence	<i>misL</i>
Secretion system	<i>invA, invB, invC, invE, invF, invG, invH, invI, orgA, orgC, prgH, prgI, prgK, sicA, spaO, spaP, spaQ, spaR, spaS, ssaC, ssaD, ssaE, ssaG, ssaJ, ssaK, ssaL, ssaM, ssaN, ssaP, ssaR, ssaU, ssaV, ssaC, ssaB, ssaE, ssaR, sipA, sipB, sipC, sopB/sigD, pipB, sijB, sspH2</i>
Translation of the Vi polysaccharide	<i>vexA, vexB, vexC, vexD, vexE</i>
Biosynthesis of Vi antigen	<i>tvbB, tvbC, tvbD, tvbE</i>
Host cell interaction (Typhoid toxin)	<i>cdtB</i>
SPI-2 mediated secretion	<i>spiC/ssaB</i>

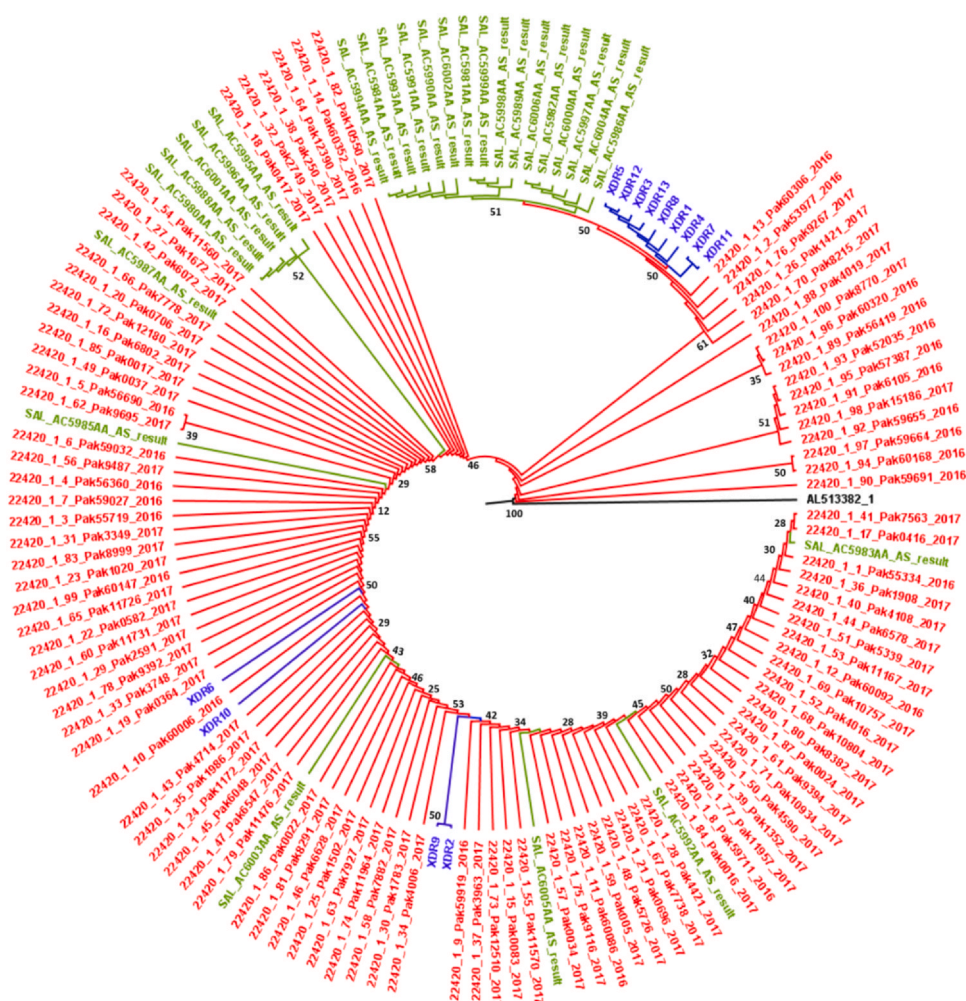


Fig. 1. The phylogenetic tree of newly sequenced 13 (Blue) Hayatabad-Peshawar XDR Typhi isolates in comparison with previously reported Pakistani XDR *S. Typhi* samples, i.e. 100 Sindh (Red) and 27 Lahore (Green) samples. The *Salmonella enterica* subsp. *Enterica* serovar *typhi* CT18 sequence (AL513382.1) was used as reference. The tree was built via REALPHY. The tree topology was visualized and annotated via FigTree.v1.4.4.

with the previously reported XDR *S. Typhi* samples from Sindh, Pakistan. The phylogenetic analysis highlighted no significant gene flow barriers among the new and previously reported Pakistani XDR *S. Typhi* isolates. This is evident with low bootstrap support for major clades of phylogenetic tree (Fig. 1). This suggests possible epidemiological linkages among these Pakistani XDR *S. Typhi* isolates and the cases emerged in Peshawar are conceivably be a part of the same outbreak that emerged earlier in the Sindh and Punjab provinces of Pakistan [7,13]. Additionally, the core and pan genome Neighbor joining (NJ) phylogenetic tree analysis also depicted similar trend of clustering (Fig. S5). This unveiled low levels of genetic differentiation and homogeneous genetic composition of all the XDR samples acquired from Pakistani regions. These isolates might have been originated from a single ancestral strain that acquired the XDR phenotype and have strong epidemiological connection.

Plasmids characterization

The WGS data of Peshawar XDR *S. Typhi* were analyzed using the PlasmidSPADES and further scanned against the entire PLSDb plasmids repository using BLASTn. The WGS data alignment of XDR1, XDR3, and XDR4 samples resulted in $\geq 90\%$ sequence coverage of the pMAC (p2ATCC19606) plasmid by each of these samples sequences. Likewise, $\geq 88\%$ sequence coverage of the pAbTS2 (p1ATCC19606) plasmid was identified by the WGS sequences of XDR1, XDR3, and

XDR4 samples. There are no published reports about characterization of these plasmids in *S. Typhi* species. Additional analyses were therefore pursued to check the existence of these plasmid sequences among the publically available draft and complete genome assemblies of *S. Typhi* deposited to Genbank, NCBI till June 2023. However, no significant sequence homologs of these plasmids were identified in the entire *S. Typhi* genome sequences available in Genbank NCBI. The pMAC (p2ATCC19606) and pAbTS2 (p1ATCC19606) have recently been characterized as native plasmids of *Acinetobacter baumannii* ATCC 19606 T. Both of these are reported as transmissible plasmids [51]. The *A. baumannii* ATCC 19606 T strain has been used as a reference in many laboratories because of its low virulence, amenability to genetic manipulation, and extensive antibiotic susceptibility. These two plasmids are reported to coexist in several strains of *A. baumannii* [52]. The opportunistic *A. baumannii* pathogen has been reported to be responsible worldwide for large nosocomial infection outbreaks, primarily in intensive care units. Several studies have recently been published about the high prevalence of drug-resistant *A. baumannii* in Pakistani hospitals and health care units [53,54]. The pMAC plasmid carries several genetic determinants that enable *A. baumannii* strains to show resistance to organic peroxides [55]. pMAC is a mobilizable plasmid with *oriT* located immediately upstream of *mobA*, which encodes a protein predicted to be involved in DNA conjugative mobilization. The pMAC-harboring genes have been reported to be present in other

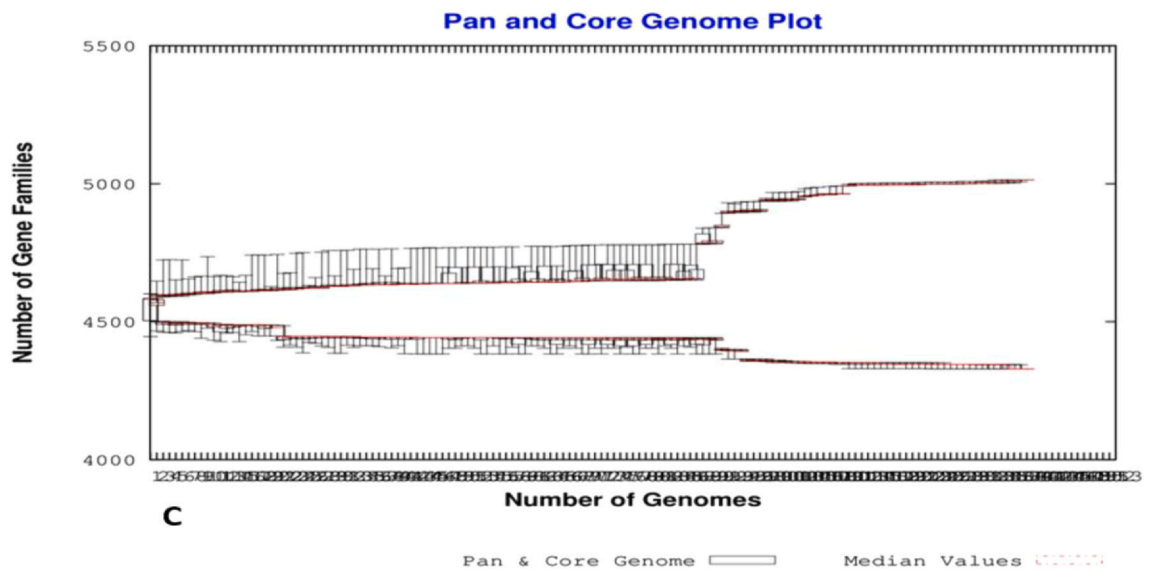
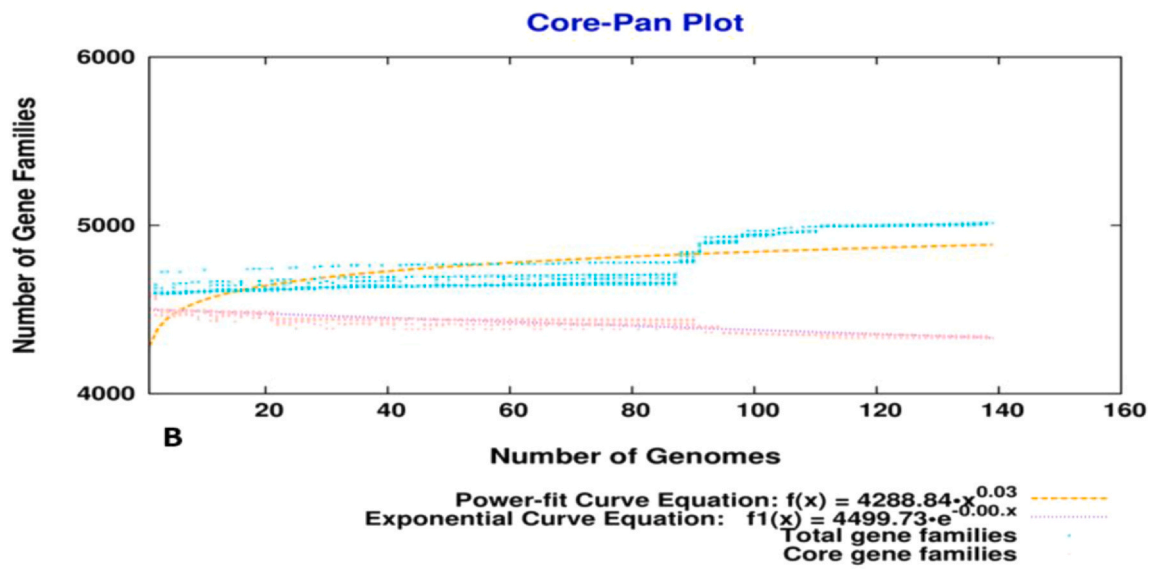
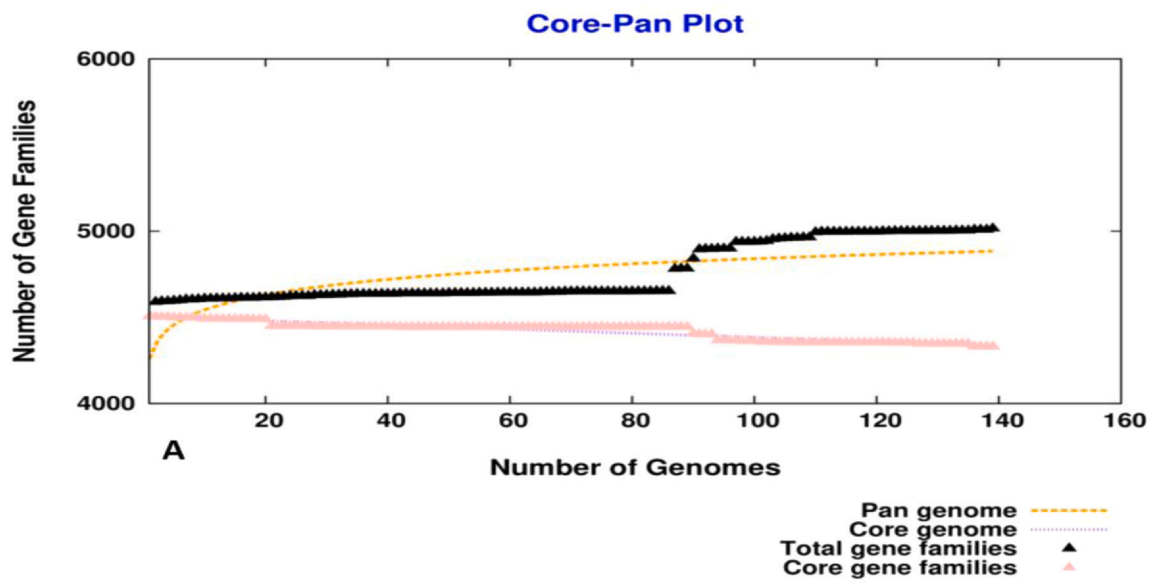


Fig. 2. The core-pan genome curve of studied *S. Typhi* genomes. (A) Total number of gene families (pangenome) and shared gene families (core genome) are plotted for a given number of genomes added sequentially. Total gene families are indicated as mustard dashed line while pink dashed line denotes core gene families. (B) The mustard dashed line indicates the least-square fit to the power law function $f(x) = a \cdot x^b$ where $a = 4288.84$, $b = 0.0264129$. The pink dashed line is the least-squares fit to the exponential decay function $f(x) = c \cdot e^{(d \cdot x)}$ where $c = 4499.73$, $d = -0.000272878$. (C) Pan- and core-genome curve of *S. Typhi* strains.

transmissible plasmids of *Acinetobacter* and *C. freundii*, which code for antimicrobial resistance. Additionally, *ohrA* and *ohrR* of pMAC are located within a region limited by perfect inverted repeats. These repeats promote the spread of *ohrA* and *ohrR* among other bacteria through different recombination mechanisms that may not involve transposition. This possibility is supported by the similarity of some regions of pMAC with *Acinetobacter* plasmids capable of spreading mercury resistance [56].

The pAbTS2 plasmid is expected to harbor Open Reading Frames (ORFs) that encode a toxin and a corresponding neutralizing antitoxin. These ORFs form a Type II Toxin-Antitoxin (TA) module, which plays a role in the maintenance and stability of the plasmid. *A. baumannii* pAbTS2 encodes the HlgB2 toxin that halts protein synthesis by acting as an mRNA ribonuclease, and therefore causes the death of daughter cells that do not inherit the pAbTS2 plasmid. HlgB2 toxin activity is neutralized by the HlgA2 antitoxin that ensures the survival of *A. baumannii* with the pAbTS2 plasmid [52]. Therefore, we speculated that XDR1, XDR3, and XDR4 *S. Typhi* samples might have acquired these plasmids from *A. baumannii*

strains during hospital-based co-infection. However, screening of these plasmids existence in a large number of *S. Typhi* clinical samples will be worthwhile to validate the findings of the current study. Exploration of the possible roles of pMAC and pAbTS2 plasmids inheritance in drug-resistant *S. Typhi* species requires additional attention.

Additionally, a partial sequence of IncQ1 plasmid was characterized for WGS data of XDR2 and XDR9 samples. The IncQ1 are small plasmids of 10–12 kb that have a well-conserved structure, and broad host range. IncQ1 plasmids replicate via strand-displacement and contain a relaxase gene, *mobA*, fused to a *repB* primase. The relaxase genes of IncQ1 plasmids are diverse and exhibit only 84% sequence homologies [57]. Besides, the ceftriaxone-resistant IncY plasmid was characterized in case of single XDR10 Peshawar sample. IncY plasmid prevalence has previously been reported in XDR *S. Typhi* samples from Sindh and Punjab regions of Pakistan [7,13]. The horizontal gene transfer (HGT) events are formerly been reported among *Acinetobacter* sp., *Pseudomonas* sp., and *Salmonella* sp. and this property of innate resistance is exclusively plasmid-borne [58].

COG Distribution

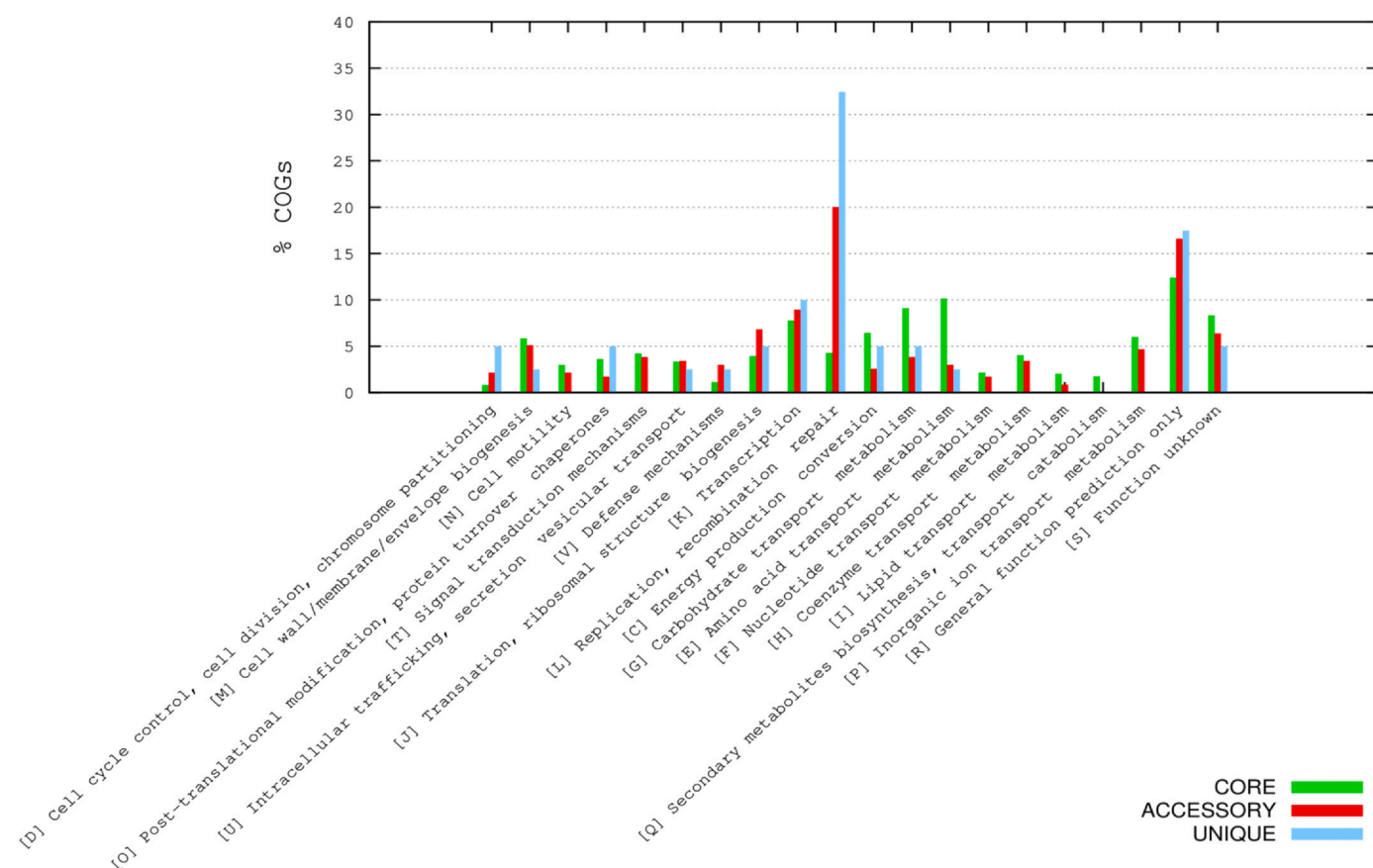


Fig. 3. COG functional distribution of pan-genome. The histogram illustrates the predicted functionality of proteins assigned to core (green), accessory (red) and unique (blue) genes. **COG categories:** [C] Energy production & conversion; [D] Cell cycle control, cell division, chromosome partitioning; [E] Amino acid transport & metabolism; [F] Nucleotide transport; [G] Carbohydrate transport & metabolism; [H] Coenzyme transport & metabolism; [I] Lipid transport & metabolism; [J] Translation, ribosomal structure & biogenesis; [K] Transcription; [L] Replication, recombination & repair; [M] Cell wall/membrane/envelope biogenesis; [N] Cell motility; [O] Post-translational modification, protein turnover & chaperones; [P] Inorganic ion transport & metabolism; [Q] Secondary metabolites biosynthesis, transport & catabolism; [R] General function prediction only; [S] Function unknown; [T] Signal transduction mechanisms; [U] Intracellular trafficking, secretion & vesicular transport; [V] Defense mechanisms.

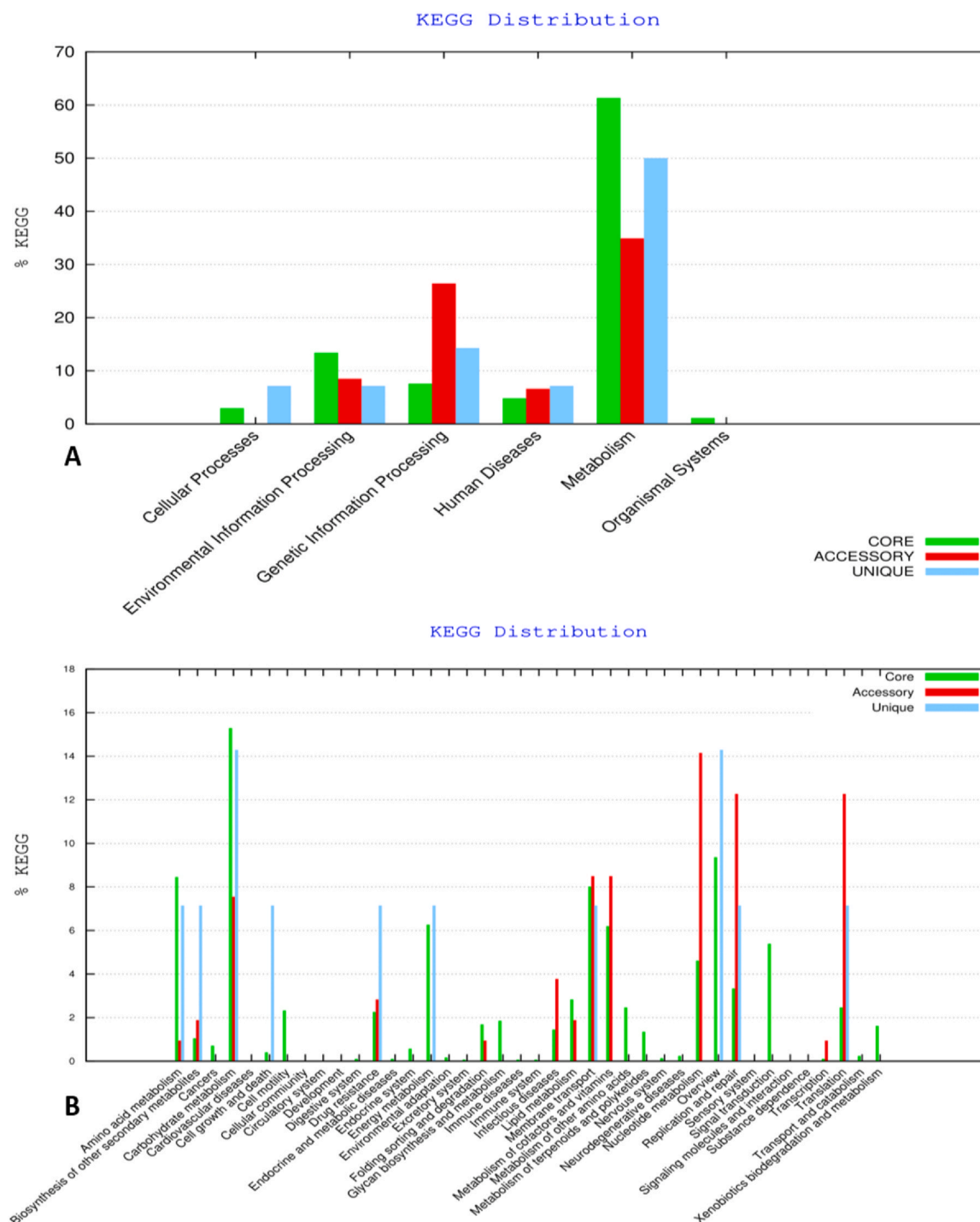


Fig. 4. KEGG functional distribution of pan-genome. **A)** The graph represent the KEGG functionality of proteins assigned to core (green), accessory (red) and unique (blue) genes. The KEGG families were broadly grouped into five category namely cellular processes, environmental information processing, genetic information processing, human diseases, metabolism, and organismal systems. **B)** Details of KEGG functional distribution of pan-genome.

Genomic data analyses of the *S. Typhi* XDR isolates from Peshawar identified the existence of the blaCTX-M-15 gene in all samples; however, the IncY plasmid partial replicon was characterized only in XDR10 strain. Besides, the IncQ1 plasmid was characterized in XDR2 and XDR9 samples. The IncQ1 plasmid carries three resistance genes; aph(6)-Id, aph(3'')-Ib (streptomycin resistant) and sul2 (sulfamethoxazole resistant also called Co-trimoxazole). The IncQ1 plasmid was previously reported in few XDR samples reported from Punjab province of Pakistan [59].

Pan and core genome analysis

A compiled dataset of 140 *S. Typhi* strains genomes, including XDR Peshawar strains, was analysed and found to exhibit close pan-genome feature. The pan-genome furnishes a comparative analysis across same species but different strains to identify species specific genomic features. The closed or open nature of pan-genome evaluates the versatility of the studied strains. The expected gene number in the pan and core-genome was calculated via curve fitting in

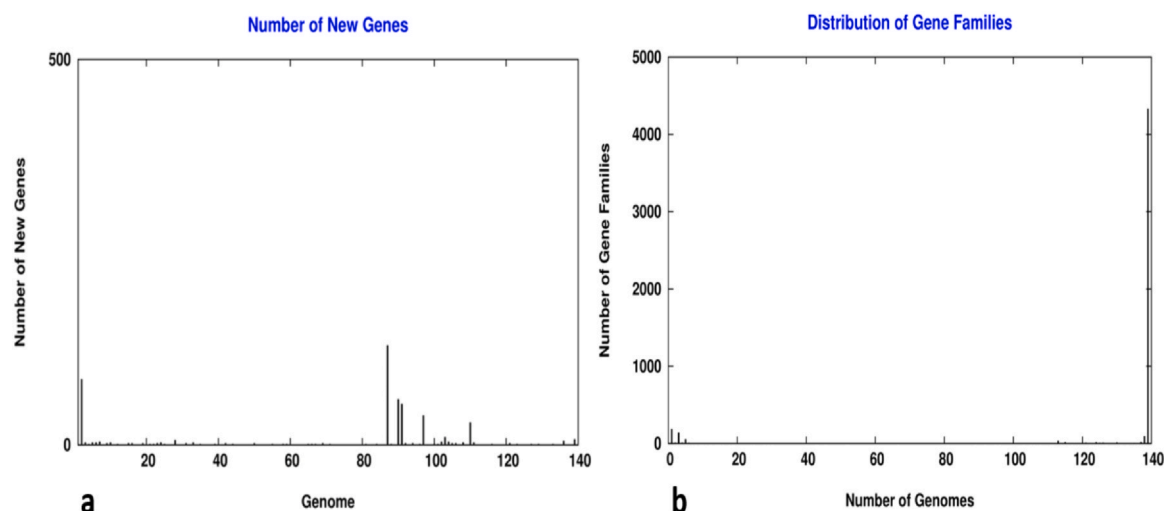


Fig. 5. (a) Number of new genes contributing to pan-genome for upon sequential addition of each genome (b) COG distribution of the number of gene families of the studied genomes.

accordance with Heaps' law [38], $Y_{\text{pan}} = A_{\text{pan}} \cdot x_{\text{pan}}^B + C_{\text{pan}}$, where y is the size of pan-genome, x is the number of genomes and A_{pan} , B_{pan} , and C_{pan} the fitting parameters and exponential equations ($Y_{\text{core}} = A_{\text{core}} \cdot e^{-B_{\text{core}} \cdot x} + C_{\text{core}}$), respectively. In this equation, B_{pan} defines whether the pan-genome is closed ($B_{\text{pan}} < 0$ or $B_{\text{pan}} > 1$) or open ($0 < B_{\text{pan}} < 1$) [60]. Total number of gene families (pan-genome) and shared gene families (core genome) are plotted for a given number of genomes added sequentially. The plot indicated a minor expansion of the pan-genome ($B_{\text{pan}} = 0.03$) while the core genome appeared to extend to almost a closed state as the addition of newer genomes failed to increment any novel gene in the existing genome (Fig. 2).

Functional distribution of genes in pan-genome

The pan-genome was further gauged for functional role of the constituent genes. The core, accessory and unique genes were examined for their functional perspective by scanning the COG and KEGG databases [39,40]. This annotation reflected that the genes are scattered in an assorted range of diverse functional categories throughout the genome. The COG distribution predicted that the core genome associated with metabolic functions like transport and metabolism of nucleotides, amino acids, coenzyme, carbohydrates, secondary metabolites biosynthesis, lipids and inorganic ions, energy production, conversion, transport and catabolism [61]. These categories had some representation in the accessory genome as well and some genes were implicated in translation and ribosomal structure mechanism. This analysis suggests that the majority of the core genes are essential for cell survival and necessary for the basic activity of the species. Nearly genes in the accessory and unique genome were primarily involved in basic cellular processes of 'information storage and processing', namely translation, ribosomal structure and biogenesis, replication, recombination, repair, and transcription. Though observed in the core genome, these genes were highly represented in the accessory genome signifying inter-species variations. The 'cellular processes and signalling' category was enriched among the core genes and depleted in the accessory genome (Fig. 3). The KEGG functional distribution revealed similar overall division of core genes associated with "metabolisms" that accounted for a huge part in core compared to accessory and unique genomes (Fig. 4). Genes were mainly associated with 'carbohydrate metabolism', 'overview', 'amino acid metabolism' and 'energy metabolism'. A sizeable gene proportion in unique and accessory genome, compared to the core-genome, participated in 'human diseases' related to 'infectious diseases' and 'drug resistance'.

Remarkably, 67 genes in the core-genome were associated with pathway 'drug resistances: antimicrobial' comprising 36 ko01503: cationic antimicrobial peptide (CAMP) resistance genes, 22 ko01501: beta-lactam resistance genes, and 9 ko01502: vancomycin resistance genes. This suggests that antibiotic resistance genotype plays a vital role in determining *S. Typhi* susceptibility. Overall 2708, 81 and 12 genes in core, accessory and unique-genome, respectively were assigned to a KEGG category. The KEGG functional distribution of pan-genome is summarized in File S4. The number of new genes contributing to pan-genome and COG distribution of gene families of the studied genomes are shown in Fig. 5.

The overall analysis unveiled closed pan-genome for Peshawar XDR *S. typhi* isolates, which is congruent to the finding of former studies [62]. The closed pangenome represents a theoretical model in which the set of genes in a bacterial species is finite and completely defined, meaning that no new genes are discovered when additional genomes are sequenced. This is in contrast to an open pangenome, where the number of genes in a bacterial species is not finite and new genes are continually being discovered as more genomes are sequenced. However, it is currently believed that most bacterial species have an open pangenome due to high levels of genetic diversity and horizontal gene transfer [63].

The current genomic study of XDR *S. Typhi* isolates from Peshawar, i.e., northwestern part of Pakistan, is based on a limited sample size. However, future exploration in the context of this study will be worthy to provide additional insight into the rate of evolution and clonal expansion of the XDR *S. Typhi* across Pakistan.

Conclusion

S. Typhi continues to pose a major challenge in treatment as it causes invasive infections and acquires antimicrobial resistance (AMR) genes to become non-susceptible to available drugs. Genomic surveillance of clinical isolates is indispensable for building a roadmap for the rational use of antibiotics and prevention strategies against enteric fever. WGS data comparison of XDR *S. Typhi* isolates revealed a uniform genotypic background of XDR typhoid-causing strains across the widespread regions of Pakistan. Genomic data indicated the existence of partial replicons of IncY, IncQ1, pMAC and pAbTS2 plasmids among the Peshawar XDR *S. Typhi* strains. Additional validation of these plasmids inheritance among the clinical *S. Typhi* isolates might be promising in the future concerning the emergence of XDR typhoid. The genome based phylogenetic analysis revealed low level of genetic diversity and persistent

circulation of XDR *S. Typhi* clone across the wide-spread region of Pakistan. The pan-genome analysis revealed the characteristics of gene pool shared by clinical *S. Typhi* isolates and closed pangenome was characterized for the new XDR Peshawar isolates focused in current study. The study highlights the importance of WGS-based resistance gene screening for tracking of AMR mechanisms in *S. Typhi* and contribute to the battle against the expanding AMR threat worldwide. The genomic data generated in the current study may utilize in future molecular epidemiological studies about clinical *S. Typhi* isolates.

Declaration of competing interest

The authors declare that no competing financial interests/ personal relationships exist in relevancy to the above-subject submitted manuscript.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jiph.2023.12.002.

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